Yubo Cui

Tel: +86 139 0469 0115 (WeChat) | Email: yubocui02@gmail.com

Homepage | Google Scholar | GitHub

Education

Harbin Institute of Technology, Shenzhen *Sep 2024 – Present*
Master of Computer Science and Technology
GPA: 3.52/4.00 | Advisor: Prof. Zheng Zhang

Harbin Institute of Technology, Weihai *Sep 2020 – Jun 2024*
Bachelor of Software Engineering
GPA: 92.32/100 | Rank: 3/173 (Top 2%)

Objective

Seeking a Ph.D. position starting in Fall 2027, focusing on **trustworthy multimodal foundation models and agents**.

Publications and Manuscripts

AGFT: Alignment-Guided Fine-Tuning for Zero-Shot Adversarial Robustness of Vision-Language Models.
Yubo Cui, Xianchao Guan, Zijun Xiong, and Zheng Zhang.
CVPR, 2026. [Paper] [arXiv] [Code]

Textual Response-Aware Guidance for Zero-Shot Adversarial Robustness of Vision-Language Models.
Yubo Cui, Xianchao Guan, and Zheng Zhang.
arXiv preprint, 2026. [arXiv]

Breaking Information Islands in Sparse Tuning via Small-World Connectivity.
Xianchao Guan, Zijun Xiong, Yifeng Wang, **Yubo Cui**, Yaowei Wang, Guosen Xie, Xin Li, and Zheng Zhang.
NeurIPS, 2026 (**Spotlight**). [Paper] [arXiv] [Code]

Frozen Reservoirs Enable Full-Rank Adaptation with Vector-Level Trainable Parameters.
Xianchao Guan, Zijun Xiong, Yifeng Wang, **Yubo Cui**, Yaowei Wang, Xin Li, and Zheng Zhang.
Under review.

Research Experience

My research interests lie in **AI security**. During my master's studies, I have primarily focused on **adversarial robustness of vision-language models**, while also working on parameter-efficient fine-tuning (PEFT).

Alignment-Guided Adversarial Fine-Tuning for Zero-Shot Robust VLMs *Aug 2025 – Mar 2026*

- **Motivation:** Hard-label supervision in classification-guided adversarial fine-tuning neglects relative image–text similarities, disrupting pre-trained cross-modal alignment and degrading zero-shot generalization.
- **Method:** Proposed AGFT, an **alignment-guided** method using the pre-trained model's probabilistic predictions as soft targets to align adversarial visual features with text features. Introduced distribution consistency calibration with temperature scaling to preserve pre-trained cross-modal semantic structure.
- **Results:** Outperformed prior methods in average clean and robust accuracy across 15 benchmarks.
- **Output:** First-author paper at **CVPR 2026**.

Textual Response-Aware Guidance for Zero-Shot Adversarial Robustness of VLMs *Mar 2026 – Present*

- **Motivation:** Pre-trained guidance helps preserve zero-shot performance during adversarial fine-tuning. Yet feature matching and prediction distribution matching have complementary limitations in preserving image–text relationships across potential semantic variations, degrading zero-shot performance.
- **Analysis:** Expressed both forms of guidance as quadratic penalties on visual feature drift, locally approximating distribution matching. Feature matching covers all directions without semantic emphasis; distribution matching favors semantic directions but is limited by candidate texts and probabilities.

- **Method:** Proposed TRAG, using the frozen text encoder's responses to local input embedding changes to guide adversarial fine-tuning. At known text anchors, these responses provide directional references in the shared feature space for potential semantic variations beyond the given texts. To preserve relative image–text similarities, TRAG penalizes drift along **response directions** in proportion to response magnitudes. Under a local response model and a fixed budget, this allocation minimizes average worst-case similarity error. Pre-trained **image–anchor relevance** on clean images weights these constraints for each image.
- **Results:** Achieved state-of-the-art clean and robust results across architectures, tasks, and threat models, with robustness transfer to large VLMs. Analyses showed TRAG helps retain pre-trained image–text relationships.
- **Output:** First-author *arXiv preprint*, 2026.

Parameter-Efficient Fine-Tuning: Sparse and Full-Rank Adaptation *Sep 2025 – Apr 2026*

- **Sparse tuning:** Sparse updates can isolate feature groups. Used a fixed scattering permutation to connect distant groups, creating small-world connectivity for task-specific information mixing.
- **Full-rank adaptation:** Used vector-level trainable parameters to modulate fixed pathways in frozen reservoirs.
- **Contribution:** Participated in method design, code implementation, and experiments; co-authored a NeurIPS 2026 Spotlight paper and a manuscript under review.

Honors and Awards

Master's First-Class Scholarship	2025
Master's Special Grade Scholarship	2024
Outstanding Graduate of Shandong Province (Top 6%)	2024
Outstanding Graduate of Harbin Institute of Technology (Top 10%)	2024
National Scholarship (Top 1%)	2022
COMAP's Interdisciplinary Contest in Modeling, Meritorious Winner (Top 6%)	2022